Exogenously Incomplete Markets

Advanced Macroeconomics

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Models with Heterogeneous Agents

- Economic agents are heterogeneous along important dimensions:
  1. Age
  2. Productivity
  3. Wealth
  4. Education, Gender, Marital Status
  5. Location, Information, Beliefs
- Number of agents: two, several (OLG, networks), or a continuum
- Ex-ante or ex-post heterogeneity; finite or infinite horizon
- **This class:** continuum of agents (here, households, next, firms), mostly ex-post heterogeneity, infinite horizon, discrete time.

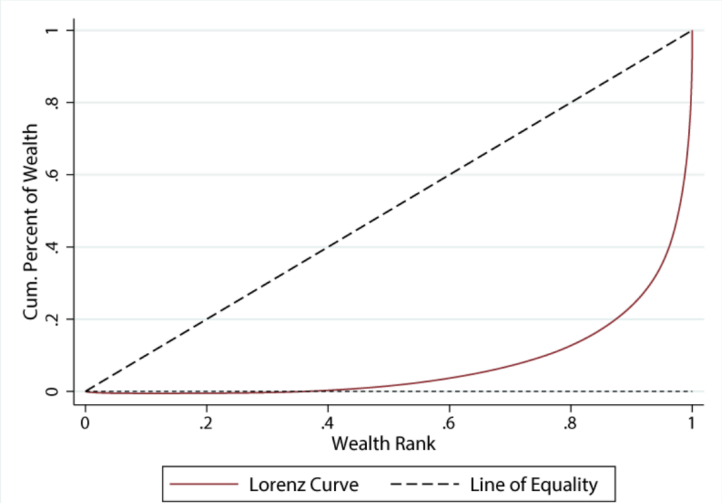
When is Heterogeneity Important?

- Questions intrinsically about heterogeneity and redistribution:
  1. Why are income and wealth so unequally distributed?
  2. What are the forces that lead to the concentration of economic activity in a few very large firms?
  3. What are the consequences of changes in welfare programs?
  4. What are the redistributive effects of inflation?
  5. What are the effects of temporary tax cuts or monetary policy?
- Is heterogeneity important for macroeconomic outcomes? Aggregation bias:
  - Is there a trade-off between inequality and economic growth? Does inequality matter for business cycle?

A Warning: Exact Aggregation

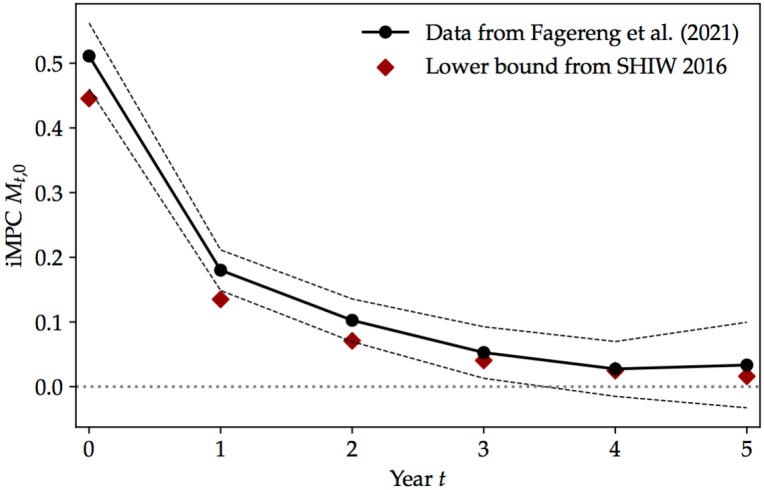
- Not every question requires a model with heterogeneous agents.
- Even when it does, we can sometimes obtain an exact aggregation result:
  1. Irrelevance result: Caplin and Spulber (1987).
  2. Exponential structure of distribution: Calvo pricing.
  3. Quasi-representative agent: Angeletos (2007).
  4. Block-recursivity: Menzio and Shi (2010).
- Subtle issue of how to compare a representative-agent and a heterogeneous-agent model.

Income and Wealth Distribution Very Unequal



(US wealth Lorenz curve, giving percent of total wealth held by poorest x% of population, from Aladangady and Forde 2021, corresponding to Gini of 0.85, which is gap between curve and 45-deg line times 2)

MPCs Are Far Higher Than Zero



(from Auclert, Rognlie, Straub 2018, "intertemporal MPCs" out of year-0 transfer)

Why Market Incompleteness?

- With complete markets: full consumption insurance, no inequality in consumption despite income shocks.
- With incomplete markets: shocks affect consumption. A (typically unique) steady state with wealth inequality emerges.
- In addition, we obtain precautionary savings: people save because they worry about future shocks.
- Key contributions:
  - Bewley (undated)
  - Imrohoroglu (1989): PE, welfare cost of business cycles
  - Huggett (1993): GE, steady state — asset pricing
  - Aiyagari (1994): GE, steady state — aggregate precautionary saving

Outline

1. **Part I:** Partial equilibrium
   - The household problem, stationary distribution
2. **Part II:** The Aiyagari model (general equilibrium, production economy)
   - Adding production, recursive competitive equilibrium, existence
3. **Part III:** Quantitative implications and calibration
   - The wealth distribution puzzle, preference heterogeneity

Environment: Building Blocks

- Continuum of measure 1 of households
- Individuals are subject to idiosyncratic shocks to their income. Idiosyncratic shocks uncorrelated across agents.
- Exogenously incomplete markets: only asset is a one-period riskless bond
- There is a borrowing constraint
- Prices are known; no aggregate uncertainty

The Household Problem

- **Preferences:** $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$, with $\beta < 1$.
- **Income process:** $y_t \in Y = \{y_1, \ldots, y_N\}$, first-order Markov with transitions $\pi(y'|y)$.
- **Budget constraint:** $c_t + q_t a_{t+1} = y_t + a_t$
- **Borrowing constraint:** $a_{t+1} \geq \underline{a}$ for all t

Recursive Formulation

- Individual state: (a, y), with a endogenous and y exogenous.
- Household problem in recursive form:

$$V(a, y) = \max_{c \geq 0,\, a' \geq \underline{a}} u(c) + \beta \sum_{y' \in Y} \pi(y'|y)\, V(a', y')$$

$$\text{s.t. } c + qa' = y + a$$

- $V(a, y)$ is the value function; $c(a, y)$ and $a'(a, y)$ are the associated policy functions.
- First-order condition: $u'(c) \geq \beta \sum_{y'} \pi(y'|y)\, V_a(a', y')$
- Envelope condition: $V_a(a, y) = (1/q)\, u'(c)$

How Do We Represent Distribution of Households?

- Last slide problem faced by an individual household
- Here we have a continuum of such households
  - Due to incomplete markets they don't all look identical ex post!
- This is a heterogeneous-agent economy
  - with a distribution of households across the two states, exogenous y and endogenous assets a
- The distribution is represented by a measure μ
- If finitely many y, then we can define $\mu(y, \mathcal{A})$ separately for each y, as a measure on subsets $\mathcal{A}$ of the asset space

The Stationary Distribution

- The law of motion of μ is:

$$\mu'(y', \mathcal{A}) = \sum_y \mu\left(y, (a')^{-1}(y, \mathcal{A})\right) \cdot \pi(y'|y)$$

  where π is transition probability and $(a')^{-1}$ is inverse of policy a'

- Measure of $\mathcal{A}$ tomorrow is sum of measures today that send you there tomorrow

- In a stationary equilibrium, $\mu(y, \mathcal{A})$ satisfies:

$$\mu(y', \mathcal{A}) = \sum_y \mu\left(y, (a')^{-1}(y, \mathcal{A})\right) \cdot \pi(y'|y)$$

- Intuition: in a stationary equilibrium, individuals jump up and down between individual states, but the overall distribution looks always the same

General Equilibrium vs. Partial Equilibrium

- Can we find a stationary equilibrium *without* imposing market clearing? Yes!
- This is useful:
  - When q changes (within a range), the distribution changes smoothly.
  - When q is too low, assets drift to infinity: no stationary distribution.
- For many applications (e.g., inequality analysis), partial equilibrium suffices.

From Endowment to Production Economy

- To study growth and business cycles, more natural to embed incomplete markets in a production economy: Aiyagari (1994).
- Gives a theory of how income risk leads to wealth inequality.
- Adding aggregate shocks: welfare cost of business cycles, etc.
- **Notation:** we now use ϵ for labor endowment and r, w for factor prices.

The Aiyagari Model: Primitives

- Continuum of measure 1 of households.
- Labor endowment $\epsilon \in \mathcal{E}$, first-order Markov with $\pi(\epsilon'|\epsilon)$.
- Labor income: $w_t \epsilon_t$.
- LLN holds: $\pi(\epsilon'|\epsilon)$ gives the deterministic fraction of the population with each transition.
- Unique stationary distribution Π over ϵ. Total labor:

$$\bar{L} = \sum_{\epsilon \in \mathcal{E}} \epsilon \, \Pi(\epsilon)$$

- Idiosyncratic shocks, but no aggregate shocks.

Preferences and Constraints

- **Preferences:** $E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$, with $\beta < 1$.
- **Budget constraint:** $c_t + a_{t+1} = w_t \epsilon_t + (1 + r_t) a_t$
- **Borrowing constraint:** $a_{t+1} \geq 0$.
- **Initial conditions:** (a_0, ϵ_0) for each agent; population measure $\mu_0(a_0, \epsilon_0)$.
- **Allocation:** $\{c_t(a_0, \epsilon^t), a_{t+1}(a_0, \epsilon^t)\}_{t=0}^{\infty}$.

Technology

- **Production:** $Y_t = F(K_t, L_t)$, CRS, concave in both arguments.
- **Depreciation:** capital depreciates at rate $\delta < 1$.
- **Aggregate resource constraint:** $C_t + K_{t+1} - (1-\delta)K_t = F(K_t, L_t)$
- The only asset in the economy is capital.
- **Factor prices from firm optimization:**

$$r_t = F_K(K_t, L_t) - \delta$$
$$w_t = F_L(K_t, L_t)$$

Recursive Equilibrium: State Spaces

- Individual state: (a, ϵ)
- Aggregate state: $\mu(a, \epsilon)$
- $A = [0, \infty)$: set of possible asset holdings
- $\mathcal{E}$: set of labor endowment realizations
- $\mathcal{P}(\mathcal{E})$: power set of $\mathcal{E}$
- $\mathcal{B}(A)$: Borel σ-algebra over A
- $Z = A \times \mathcal{E}$, $\mathcal{B}(Z) = \mathcal{P}(\mathcal{E}) \times \mathcal{B}(A)$
- $\mathcal{M}$: set of probability measures on $(Z, \mathcal{B}(Z))$

Recursive Household and Firm Problems

- **Household:**

$$v(a,\epsilon;\mu) = \max_{c\geq 0,\, a'\geq 0} u(c) + \beta \sum_{\epsilon'\in\mathcal{E}} \pi(\epsilon'|\epsilon)\, v(a',\epsilon';\mu')$$

$$\text{s.t.} \quad c + a' = w(\mu)\epsilon + (1 + r(\mu))a$$

$$\mu' = H(\mu)$$

where $H : \mathcal{M} \to \mathcal{M}$ is the aggregate law of motion.

- **Firm:** $\max_{k,l\geq 0} F(k,l) - (r(\mu)+\delta)k - w(\mu)l$

- Policy functions: $K(\mu)$, $L(\mu)$.

Recursive Competitive Equilibrium

Definition: A RCE is a value function $v : Z \times \mathcal{M} \to \mathcal{R}$, policy functions $a' : Z \times \mathcal{M} \to \mathcal{R}_+$, $c : Z \times \mathcal{M} \to \mathcal{R}_+$, firm policies $K, L : \mathcal{M} \to \mathcal{R}_+$, pricing functions $r : \mathcal{M} \to \mathcal{R}$, $w : \mathcal{M} \to \mathcal{R}_+$, and LOM $H : \mathcal{M} \to \mathcal{M}$, such that:

(1) given prices, v satisfies the Bellman equation; a', c are the policy functions.

(2) Factor prices: $r(\mu) = F_K(K(\mu), L(\mu)) - \delta, \quad w(\mu) = F_L(K(\mu), L(\mu))$.

(3) Markets clear for all $\mu \in \mathcal{M}$:

$$K(H(\mu)) = \int a'(a, \epsilon; \mu)\, d\mu, \quad L(\mu) = \int \epsilon\, d\mu$$

$$\int [c(a, \epsilon; \mu) + a'(a, \epsilon; \mu)]\, d\mu = F(K(\mu), L(\mu)) + (1 - \delta)K(\mu)$$

(4) The LOM H is generated by a' and π.

Stationary Recursive Competitive Equilibrium

Definition: A stationary RCE is a value function $v : Z \to \mathcal{R}$, policy functions $a' : Z \to \mathcal{R}_+$, $c : Z \to \mathcal{R}_+$, aggregates K, L, prices r, w, and a measure $\mu \in \mathcal{M}$, such that:

(1) given prices v satisfies the Bellman equation; a', c attain the maximum, given r, w.

(2) $r = F_K(K, L) - \delta, \quad w = F_L(K, L)$.

(3) Markets clear:

$$K = \int a'(a, \epsilon)\, d\mu, \qquad L = \int \epsilon\, d\mu$$

$$\int [c(a, \epsilon) + a'(a, \epsilon)]\, d\mu = F(K, L) + (1 - \delta)K$$

(4) μ is stationary: for all $(\mathcal{A}, E) \in \mathcal{B}(Z)$, $\mu(\mathcal{A}, E) = \int Q((a, \epsilon), (\mathcal{A}, E))\, \mu(da \times d\epsilon)$, where Q is the transition function induced by π and a'.

Note: r and w are constant because the distribution does not change.

Simplifying Observations: Household Side

- Steady-state A depends on r and w.
- Under two assumptions–CRRA utility and zero borrowing constraint–A scales linearly in w:

$$A(r, w) = a(r)wL.$$

- Intuition: scaling w scales all policies optimally [prove this?]
- $a(r)$ continuous, increasing (for CRRA) on $r \in [-\delta, 1/\beta - 1)$

Simplifying Observations: Production Side

- Assuming CRTS production function, continuous and strictly concave, write:

$$f(k) \equiv F(k, 1)$$

- This is also strictly concave, and

$$f'(K/L) = F_K(K/L, 1) = F_K(K, L) = r + \delta$$

- Capital-labor ratio $k \equiv K/L$ solves

$$f'(k) = r + \delta$$

- Implicitly defined function $k(r)$ (declining function)

Simplifying Observations: Production Side Continued

- We also have from the firm first order conditions:

$$w = F_L(K, L) = \frac{F(K, L) - F_K(K, L)K}{L} = f(K/L) - f'(K/L)(K/L)$$

- Given $k(r)$, can substitute in to get $w(r)$:

$$w(r) \equiv f(k(r)) - f'(k(r))(k(r))$$

- Can show this is strictly declining in r as well:

$$w'(r) = -f''(k)k'(r) < 0$$

Special Case: Cobb-Douglas Technology

- Assume:

$$Y = F(K, L) = K^{\alpha} L^{1-\alpha}.$$

- Marginal products:

$$r + \delta = F_K = \alpha \frac{Y}{K}, \quad w = F_L = (1-\alpha)\frac{Y}{L}.$$

- Wage-to-interest ratio:

$$\frac{w}{r+\delta} = \frac{1-\alpha}{\alpha}\frac{K}{L}.$$

- So:

$$\frac{k(r)}{w(r)} = \frac{\alpha}{1-\alpha}\frac{1}{r+\delta}$$

- Closed-form solution for asset supply vs. labor income, unambiguously decreasing

Equilibrium: One equation in one unknown

- Key equilibrium condition is $A = K$, which becomes:

$$a(r)w(r)L = k(r)L.$$

- Dividing by L and rearranging:

$$\underbrace{a(r)}_{\textbf{Asset demand}\text{ relative to labor income}} = \underbrace{\frac{k(r)}{w(r)}}_{\textbf{Asset supply}\text{ relative to labor income}}$$

- Existence boils down to one equation in one unknown:
  - Left-hand side is likely increasing in r.
  - Right-hand side is likely decreasing in r (if numerator dominates denominator).

The Aiyagari Diagram

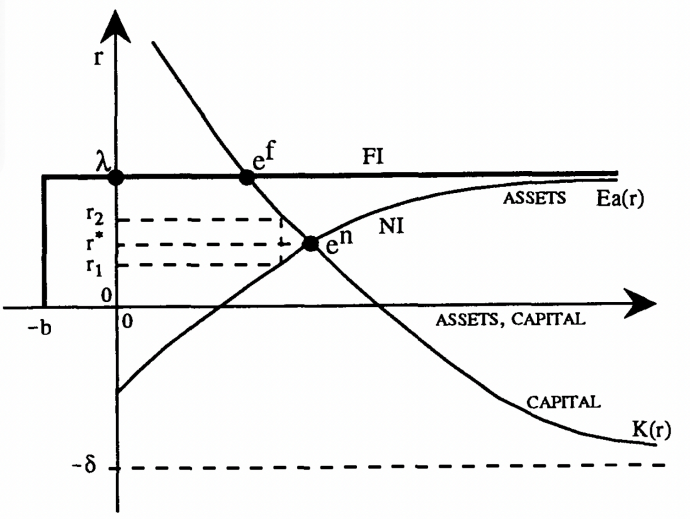


Figure 1: Source: Aiyagari (1994)

Existence

- From assumptions on F, $K(r)$ is continuous and strictly decreasing on $r \in (-\delta, \infty)$, with $\lim_{r \to -\delta} K(r) = \infty$ and $\lim_{r \to \infty} K(r) = 0$.
- $Ea(r)$ is well defined and continuous on $r \in (-\delta, 1/\beta - 1)$.
- If $\lim_{r \to 1/\beta - 1} Ea(r) > K(1/\beta - 1)$, then by the intermediate value theorem there exists r^* such that
$$K(r^*) = Ea(r^*)$$
and a stationary RCE exists.
- Uniqueness? Likely but not guaranteed.
- Stability? Open question.

Calibration Strategy

- **Frequency:** quarterly.
- **Income process:**
  - Individual labor income follows: $\log z_t = \rho \log z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$
  - Calibrated to match the U.S. pre-tax income process:
    - Annual persistence: roughly $\rho = 0.91$
    - Cross-sectional standard deviation: roughly $\sigma = 0.92$
  - Mean income normalized to 1.
- **Preferences:** CRRA utility, $u(c) = c^{1-\sigma}/(1-\sigma)$.
- **Remaining free parameter:** discount factor β, pinned down by a target. But which target?

Challenge I: The Wealth Distribution

- Asset inequality in the data is huge:
  - Gini coefficient: ≈ 0.8 for wealth (≈ 0.4 for earnings).
  - Most wealth is held by a few rich households.
  - Many households at zero or negative net worth.
- A large literature focuses on matching the top of the wealth distribution.
- Typical calibration strategy: choose β to match aggregate assets ($\approx$ 500% of GDP).

The Wealth Distribution: A Problem

The distribution of wealth							
	% of wealth held by top					Fraction with wealth < 0	Gini coefficient
	1%	5%	10%	20%	30%		
$\underline{b} = 0$ model	3%	11%	20%	35%	47%	0%	0.26
$\underline{b} = -2.4$ model	3%	13%	23%	39%	52%	0.5%	0.33
Data	30%	51%	64%	79%	88%	11%	0.79

The baseline model generates far too little wealth concentration.

Matching the Top: What Is Missing?

- We need the rich to save more. Different possibilities:
  - Bigger income shocks (Castañeda, Díaz-Giménez, and Ríos-Rull, 1993).
  - Difference in discount factors: on average, rich families are patient.
  - Higher returns for the rich (Fagereng et al., 2020).
  - Higher income when poor (e.g., unemployment insurance). Why does this help?
- Related to Piketty and Zucman (2014): with random saving rates and linear decision rules, the wealth distribution becomes Pareto.
- For agents with large asset holdings, decision rules are approximately linear. Linearity follows from the fact that wealthy agents are well insured.

Challenge II: The Marginal Propensity to Consume

- A more recent literature — especially on monetary and fiscal policy (HANK models) — focuses on matching the marginal propensity to consume (MPC).
- Why? The aggregate effect of transfers, tax rebates, or interest rate changes depends critically on how much households spend out of additional income.
- Empirical evidence: income-weighted quarterly MPC $\approx$ 0.20 (much higher than a representative-agent model would predict).
- Calibration strategy: choose β to match the MPC rather than aggregate assets.

The Tension Between Wealth and MPCs

- **Key problem:** with a single β, the two targets are in conflict.
  - A β that matches aggregate assets ($\approx$ 500% of GDP) typically implies MPCs that are too low.
  - A β that matches MPCs ($\approx$ 0.20) implies too little aggregate wealth.

- **Intuition:** a patient household accumulates a lot of wealth *and* has a low MPC (well insured). An impatient household has high MPC but little wealth. A single β cannot produce both a realistic wealth stock and realistic MPCs.

Reconciliation: β-Heterogeneity

- **Idea:** introduce heterogeneity in discounting. Patient households accumulate wealth (matching the top); impatient households have high MPCs (matching the aggregate MPC).
- **Stochastic process for β:**
  - Example: Auclert, Rognlie, Straub (2025).
  - Four equispaced states, equal population shares.
  - Households redraw β with probability 1% per quarter (one "generation" $\approx$ 25 years).
- **Targets and calibration:**
  - Aggregate assets: 500% of GDP.
  - Income-weighted quarterly MPC: 0.20.
  - Calibrated quarterly $\beta(s)$: $\{0.94,\ 0.96,\ 0.98,\ 1.00\}$

Implications of β-Heterogeneity

- Why are the poor poor? Because they are impatient. Contrast with Castañeda et al. (1993), where the poor are unlucky. Policy implications differ!
- Stochastic β preferred to permanent types to avoid extreme saver/spender polarization.
- Preference heterogeneity is now standard in the HANK literature for studying fiscal and monetary policy transmission.